EuroSAT: Land Use and Land Cover Classification with Sentinel-2
Literature Review (2022–2026)

Scope and approach

EuroSAT is close to saturated: recent papers report 98–99% overall accuracy, so 2022–2026 work is judged on transfer, label efficiency and robustness rather than raw accuracy.

EuroSAT holds 27,000 labeled, geo-referenced Sentinel-2 patches in 10 classes across 13 spectral bands, and its original benchmark reached 98.57% accuracy with deep CNNs.

This review covers papers from 2022 onward that report EuroSAT results, chosen to span the main directions: self-supervised pre-training, foundation models and lightweight task-specific CNNs. Every figure below was read from the paper or its official model card. “Relation to your work” assumes your study classifies EuroSAT Sentinel-2 patches; adjust that column if your angle differs.

Comparison of papers

Fine-tuned pre-trained models reach 98.9–99.2% on EuroSAT, while the two models trained from scratch score 96–97.2%.

Table 1. Comparison of the reviewed papers on EuroSAT.

Title	Authors	Year	Dataset	Application area	Method/model	Metrics	Strengths	Limitations	Research gap	Relation to your work
Balanced Multi-Task Attention for Satellite Image Classification: A Systematic Approach to Achieving 97.23% Accuracy on EuroSAT Without Pre-Training	A. Vir	2025	EuroSAT RGB, 27,000 images (64×64); random 70/15/15 split	Land-use classification where pre-training is unavailable (proprietary sensors, novel modalities)	12-layer ResNet-style CNN (11.2M parameters) trained from scratch; Coordinate Attention and Squeeze-Excitation fused by a learnable weight ($\alpha \approx 0.57$); progressive DropBlock; class-weighted loss	Test accuracy 97.23%; Cohen's κ 0.9692; macro F1 97.12%; every class ≥94.46%	Stepwise ablation (94.30% → 95.98% → 97.23%); per-class and confusion analysis (River ← Highway errors 27 → 5); code released	Single-author preprint; RGB only; one split and one model; still 1.34 pts below the 2019 pre-trained ResNet-50 (98.57%)	Extension to multispectral bands, ensembles and other datasets is left as future work	From-scratch CNN baseline and a template for per-class error analysis on RGB EuroSAT
A Deep CNN model for Landuse Landcover Classification for 4 Band Visible and NIR Datasets	P. Chachra, A. Tiwari, M. Kumar	2025	EuroSAT, 4 bands (B2, B3, B4, B8), 27,000 images; stratified 64/16/20 train/validation/test split	Resource-limited or edge LULC monitoring (urban growth, agriculture, hazard response)	Custom 4-block CNN (≈424.8k parameters) with batch norm, dropout and global average pooling; rotation, shift, flip and zoom augmentation	Test accuracy 96%; macro precision, recall and F1 all 0.96 on 5,400 test images	Compact model using 4 of 13 bands; stratified split; per-class F1 and confusion matrix	Lowest headline accuracy of the six; the quoted RGB-only baseline (87–91%) has no citation or own experiment; NIR effect not isolated by ablation	13-band fusion (SWIR, red-edge), multi-date imagery and transfer learning are listed as future work	Lightweight RGB+NIR band-subset baseline; shows the accuracy cost of a small model
Neural Plasticity-Inspired Multimodal Foundation Model for Earth Observation	Z. Xiong, Y. Wang, F. Zhang and 7 others (incl. D. Borth, X. X. Zhu)	2024	Pre-trained on five sensors; EuroSAT evaluated as "m-eurosat", one of six classification tasks	Sensor-agnostic EO foundation model: classification, segmentation, object detection, change detection	Single ViT with a wavelength-conditioned dynamic hypernetwork that accepts any number of channels; evaluated by linear probing	Linear-probe overall accuracy on m-eurosat: 97.1% (ViT-L), 93.9% (ViT-B), from the official model card	One model for many sensors; frozen ViT-L within 0.6 pt of fully trained ConvNeXt-B (97.7%); ahead of SatMAE ViT-B (86.4%) and CROMA (90.1%) in the same table	EuroSAT is one aggregate score among many tasks, with no per-class or robustness analysis; result depends on backbone size (ViT-B 93.9%)	Per-class errors and label-efficient behaviour on EuroSAT are not reported	Reference for a frozen general-backbone versus task-specific training

Title	Authors	Year	Dataset	Application area	Method/model	Metrics	Strengths	Limitations	Research gap	Relation to your work
Mapping of Land Use and Land Cover (LULC) using EuroSAT and Transfer Learning	S. Kunwar, J. Ferdush	2024	EuroSAT RGB, 27,000 images (64×64); random 80/20 train/test split; Sentinel-2A tiles of Kreis Borken (2018–2020) for mapping	LULC scene classification and regional land-use mapping for conservation and urban planning	ViT pre-trained on ImageNet-21k and fine-tuned; compared with VGG16 and ResNet-50, with and without augmentation	Accuracy 99.19% (ViT, augmented, 20 epochs); 98.72% at 15 epochs; ResNet-50 98.48–98.52%; VGG16 97.78–98.06%	Simple, reproducible recipe (augmentation, early stopping, gradient clipping); ViT leads both CNNs; classifies a real region end to end	RGB only; one random split, no repeated runs; headline is the best of three epoch settings (98.24–99.19%); ViT trains about 3× longer than ResNet-50; map shown without an accuracy assessment	Spatially held-out evaluation and an accuracy check of the Kreis Borken map are missing	RGB transfer-learning baseline plus a patch-to-map workflow
SSL4EO-S12: A Large-Scale Multi-Modal, Multi-Temporal Dataset for Self-Supervised Learning in Earth Observation	Y. Wang, N. A. A. Braham, Z. Xiong, C. Liu, C. M. Albrecht, X. X. Zhu	2023 (arXiv 2022)	Pre-training on SSL4EO-S12 (251,079 locations, Sentinel-1/2, four seasons); EuroSAT downstream with 21,600 train / 5,400 test images	Self-supervised pre-training for EO: scene classification, land cover segmentation, change detection	MoCo-v2/v3, DINO, MAE and data2vec with ResNet-50 or ViT-S/16; linear probing and fine-tuning	EuroSAT overall accuracy: fine-tuned 99.1% (MoCo and DINO ResNet-50, data2vec ViT-S); linear probe 98.0% (MoCo ResNet-50, all bands), equal to supervised 98.0%	Open global corpus with released weights; seasonal augmentation lifts linear-probe EuroSAT from 93.1% to 96.2% (50k-patch ablation); beats ImageNet, SeCo, SEN12MS and BigEarthNet pre-training	Little polar coverage; geographic bias from cloud filtering; not strictly free of overlap; EuroSAT gains stay within about 1 pt of supervised	EuroSAT cannot separate pre-training methods; the city-held-out So2Sat-LCZ42 task scores only about 55–65%	Source of pre-trained Sentinel-2 weights and the linear-probe versus fine-tune protocol
SatMAE: Pre-training Transformers for Temporal and Multi-Spectral Satellite Imagery	Y. Cong, S. Khanna, C. Meng, P. Liu, E. Rozi, Y. He, M. Burke, D. B. Lobell, S. Ermon	2022	Pre-training on fMoW-Sentinel (712,874 training images, 13 bands); EuroSAT downstream, 27,000 13-band images	Self-supervised pre-training for temporal and multispectral imagery: land cover classification, building segmentation	Masked autoencoder with ViT-Large; spectral bands split into groups with group encodings; independent masking	EuroSAT top-1 accuracy 98.94% (98.98% with group encoding and independent masking); RGB-only 95.74%; SeCo 93.14%; GASSL 89.51%	Large gain over contrastive baselines; 13 bands beat RGB by about 3 pts; also tested on NAIP, BigEarthNet and SpaceNet	Longer token sequences strain compute; each experiment run once with no error bars; EuroSAT is a secondary check with splits taken from earlier work	Better spectral encodings and band groupings are left open	Reference for fine-tuned 13-band performance and the value of multispectral input

Accuracies are not directly comparable: splits, band sets and evaluation protocols differ across rows.

Synthesis and research gaps

With EuroSAT accuracy near 99%, the open problems are comparability, spatial generalization and the value of extra spectral bands.

- **Results are not comparable.** The papers use different splits (random 80/20, 70/15/15, 64/16/20, 21,600/5,400), band sets (RGB, RGB+NIR, 13 bands) and protocols (linear probe or fine-tuning), so headline numbers cannot be ranked directly.
- **Variance is rarely reported.** SatMAE states each experiment was run once with no error bars; the three task-specific papers each report one split.
- **No geographic hold-out.** None of the six describes one for EuroSAT. The only geographic split in these sources, So2Sat-LCZ42 in SSL4EO-S12, scores about 55–65% against 97–99% on EuroSAT, although the tasks differ.
- **The role of spectral bands is unresolved.** SatMAE gains about 3 pts from 13 bands over RGB, yet Kunwar and Ferdush reach 99.19% on RGB alone, and Chachra et al. do not ablate NIR.

A study that adds a geographic hold-out, repeated runs and a band ablation would address the last three gaps at once.

References

1. Cong, Y., Khanna, S., Meng, C., Liu, P., Rozi, E., He, Y., Burke, M., Lobell, D. B., Ermon, S. (2022). SatMAE: Pre-training Transformers for Temporal and Multi-Spectral Satellite Imagery. NeurIPS 35. Paper
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3. Xiong, Z., Wang, Y., Zhang, F., Stewart, A. J., Hanna, J., Borth, D., Papoutsis, I., Le Saux, B., Camps-Valls, G., Zhu, X. X. (2024). Neural Plasticity-Inspired Multimodal Foundation Model for Earth Observation. arXiv:2403.15356. Abstract; EuroSAT figures from the official model card
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5. Chachra, P., Tiwari, A., Kumar, M. (2025). A Deep CNN model for Landuse Landcover Classification for 4 Band Visible and NIR Datasets. ISPRS Annals X-5/W2-2025, 65–70. Paper
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7. Helber, P., Bischke, B., Dengel, A., Borth, D. (2019). EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification. IEEE JSTARS 12(7). Abstract